CLOUD & VIRTUALISATION | TEK-UP UNIVERSITY

Lab 02

Network access modes in VirtualBox

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Class

Ing1, Groupe C (evening course)

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About this report

In this lab I tested the four network modes of VirtualBox that the handout asks for: NAT, Bridged Adapter, Internal Network and Host-Only Adapter. For each mode I noted the address of my PC and of two virtual machines, then pinged between the host, the VMs and the Internet. In the last part I installed Apache on a third VM and made it reachable from Windows with port forwarding.

I worked on Red Hat Enterprise Linux 10, so a few commands differ from the handout: `ip a` instead of `ifconfig`, `httpd` instead of `apache2`, an extra firewall step to open port 80, and the newer option names of `VBoxManage dhcpserver`.

Two results I could not explain: a VM could not ping my Windows host in Bridged mode or in Host-Only mode. I explain what I tested and what I could not prove in sections 3 and 5.

1. Preparing the VMs

The handout uses two machines. VM1 is my existing `VM1-RedHat-Desktop`. For VM2 I shut VM1 down, right-clicked it, chose **Cloner**, named the copy `VM2-RedHat-Desktop`, chose a **full clone**, and told VirtualBox to reset the MAC address of every network card. A new MAC address matters, because two VMs with the same one would disturb each other on the network.

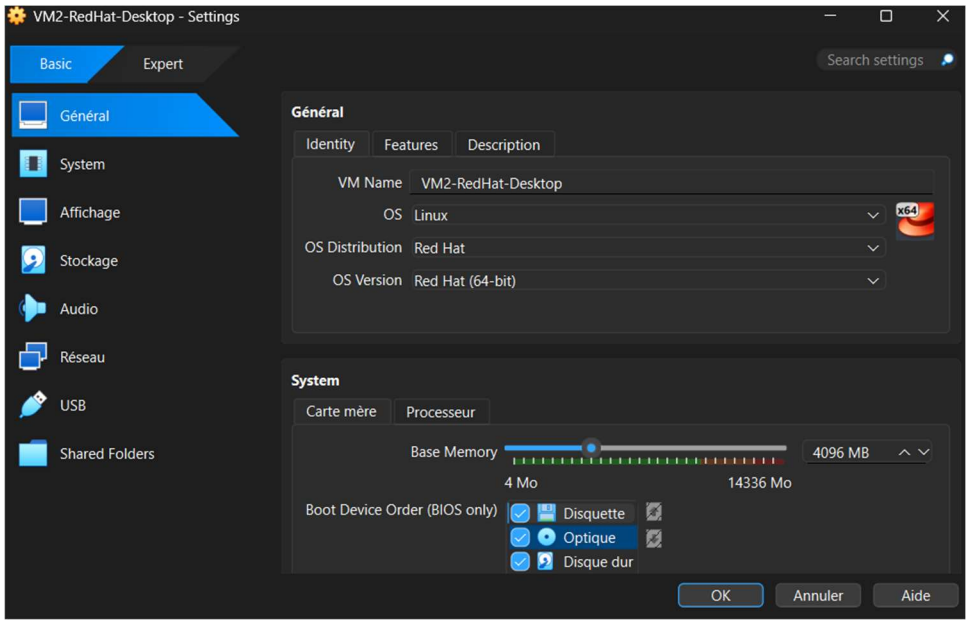


Figure 1. Settings of VM2-RedHat-Desktop right after the clone.

Later, for the Apache part, I made a third full clone called `VM3-RedHat-Desktop`.

2. NAT mode

NAT is the default mode: each VM sits behind a private network created by VirtualBox and reaches the outside through the host. I checked that both VMs were in NAT, started them and collected the addresses.

The addresses on `enp0s3` of both VMs come from `ip a`, and the host address from `ipconfig` on the Wi-Fi adapter, which is the only adapter with a default gateway. The other adapters in the list (VirtualBox Host-Only and VMware) were not used.

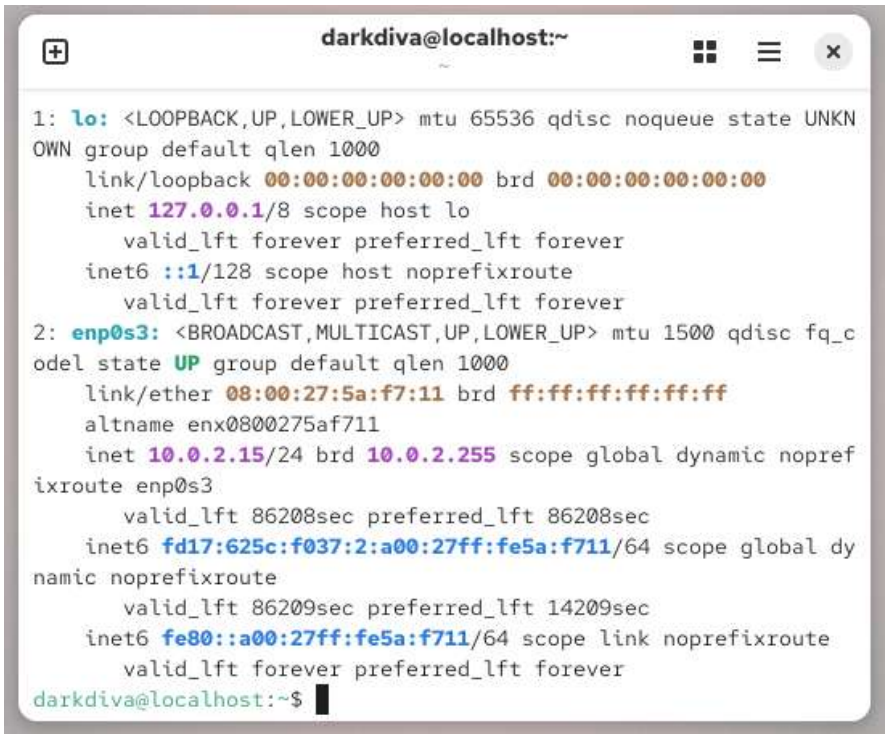


Figure 2. VM1 in NAT mode: 10.0.2.15/24 (MAC ending 5a:f7:11).

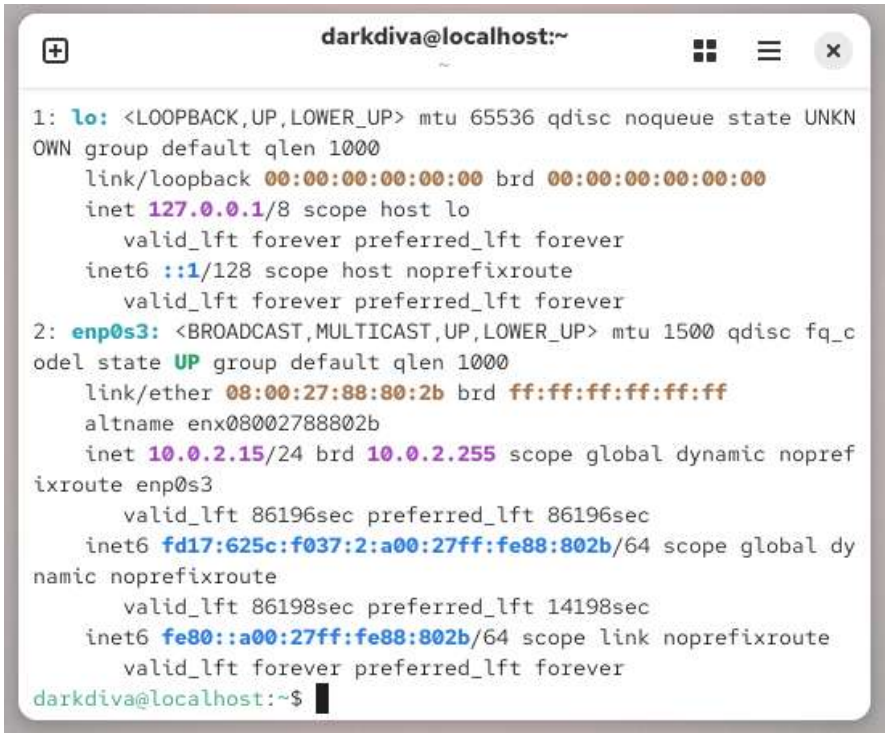


Figure 3. VM2 in NAT mode: also 10.0.2.15/24 (MAC ending 88:80:2b).

```
Carte Ethernet Ethernet 3 :

   Suffixe DNS propre à la connexion. . . :
   Adresse IPv6 de liaison locale. . . . .: fe80::f261:9a6:f0c5:8717%19
   Adresse IPv4. . . . . . . . . . . . . .: 192.168.56.1
   Masque de sous-réseau. . . . . . . . . : 255.255.255.0
   Passerelle par défaut. . . . . . . . . :

Carte réseau sans fil Connexion au réseau local* 1 :

   Statut du média. . . . . . . . . . . . : Média déconnecté
   Suffixe DNS propre à la connexion. . . :

Carte réseau sans fil Connexion au réseau local* 2 :

   Statut du média. . . . . . . . . . . . : Média déconnecté
   Suffixe DNS propre à la connexion. . . :

Carte réseau sans fil Wi-Fi :

   Suffixe DNS propre à la connexion. . . : home
   Adresse IPv6 de liaison locale. . . . .: fe80::459f:6e93:9846:6e54%12
   Adresse IPv4. . . . . . . . . . . . . .: 192.168.0.137
   Masque de sous-réseau. . . . . . . . . : 255.255.255.0
   Passerelle par défaut. . . . . . . . . : 192.168.0.1
```

Figure 4. ipconfig on Windows: Wi-Fi 192.168.0.137 and the Host-Only adapter 192.168.56.1.

Table 1. Addresses in NAT mode

Machine	IP address	Netmask
Host (Wi-Fi)	192.168.0.137	255.255.255.0
VM1	10.0.2.15	255.255.255.0 (/24)
VM2	10.0.2.15	255.255.255.0 (/24)

Ping tests

From VM1 I pinged the Internet (8.8.8.8), the host and the address of VM2. From Windows I pinged the VM.

```
darkdiva@localhost:~

       valid_lft 86209sec preferred_lft 14209sec
    inet6 fe80::a00:27ff:fe5a:f711/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
darkdiva@localhost:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=36.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=24.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=189 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=115 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 24.684/91.152/188.600/66.120 ms
darkdiva@localhost:~$ ping -c 4 192.168.0.137
PING 192.168.0.137 (192.168.0.137) 56(84) bytes of data.
64 bytes from 192.168.0.137: icmp_seq=1 ttl=64 time=0.990 ms
64 bytes from 192.168.0.137: icmp_seq=2 ttl=64 time=1.04 ms
64 bytes from 192.168.0.137: icmp_seq=3 ttl=64 time=1.06 ms
64 bytes from 192.168.0.137: icmp_seq=4 ttl=64 time=0.688 ms
```

Figure 5. VM1 to the Internet (8.8.8.8) works, then the start of the ping to the host.

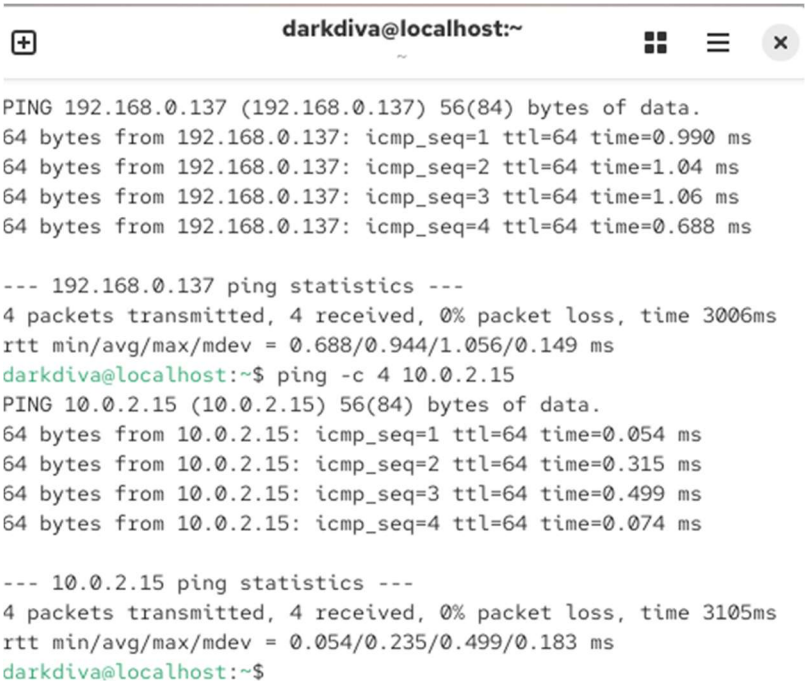


Figure 6. VM1 to the host works. The last test pings 10.0.2.15, which is VM1's own address.

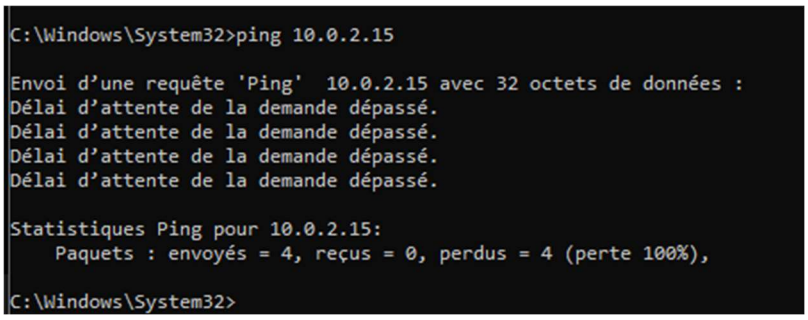


Figure 7. From Windows, ping 10.0.2.15 gets no answer (100% loss).

The VM1 to VM2 trap

Both VMs have the same address, 10.0.2.15. When VM1 pings 10.0.2.15 it gets replies in a fraction of a millisecond, but it is only talking to itself. I therefore recorded this test as **Non**: in NAT mode each VM lives in its own private network and the VMs cannot see each other.

Table 2. Connectivity in NAT mode

Source	Destination	Ping successful?
Host	VMs	Non
VMs	Host	Oui
VM1	VM2	Non (the reply came from VM1 itself)
VMs	Internet (8.8.8.8)	Oui

Conclusion: in NAT mode the VMs reach the Internet and the host, but nothing from outside can reach them, and they cannot reach each other.

3. Bridged Adapter mode

In this mode the VM connects to the same network as my PC, as if it were another computer on my Wi-Fi. I shut both VMs down and set **Adapter 1** of each to **Accès par pont**, using my Realtek Wi-Fi adapter.

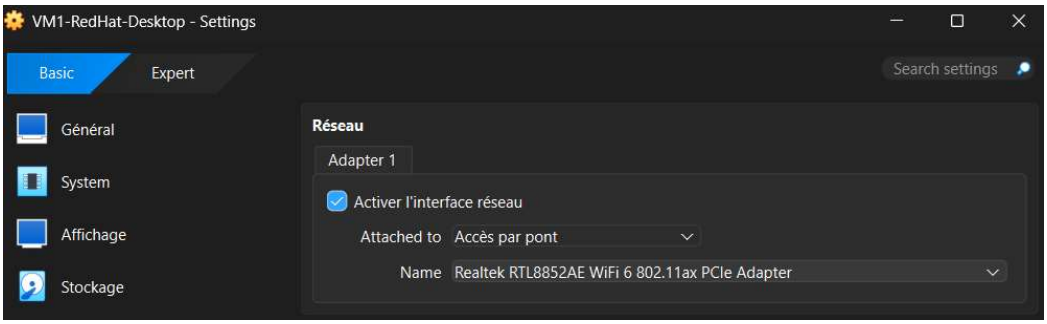


Figure 8. VM1 network settings: Bridged Adapter on the Wi-Fi card.

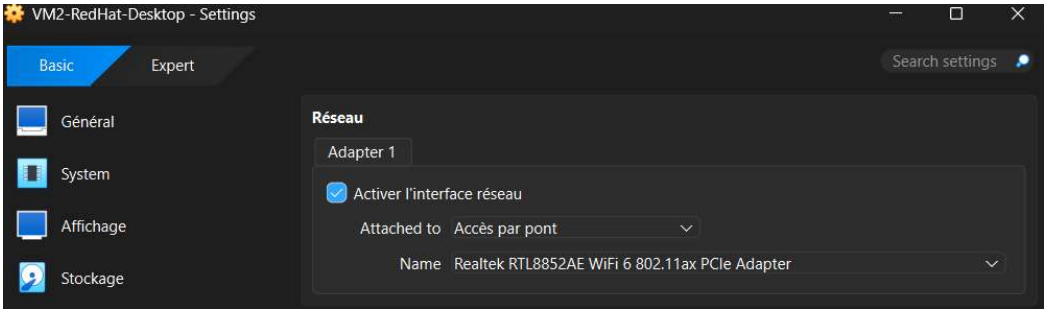


Figure 9. VM2 network settings: the same.

> **Problem I ran into**
>
> After the PC was restarted, VM1 would not start from the VirtualBox Manager, while VM2 did. Starting VM1 from the command line with `VBoxManage startvm "VM1-RedHat-Desktop"` worked, and VM1 then ran normally for the rest of the lab. I did not find the cause.

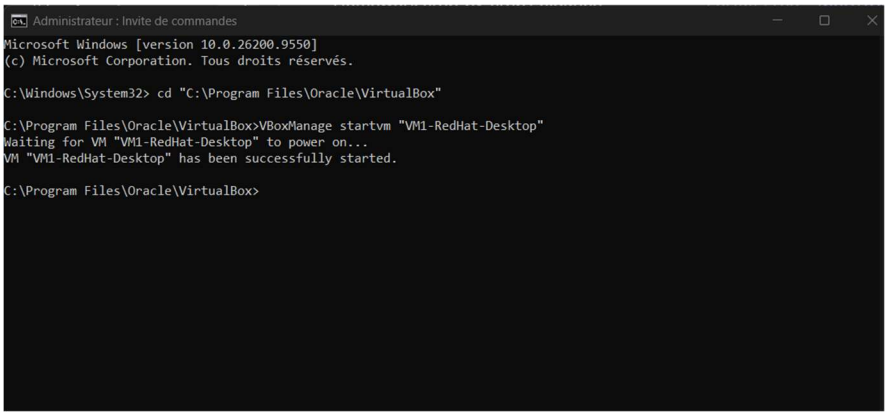


Figure 10. VM1 started with VBoxManage.

Each VM now got its own address from my home network, in the same range as the host:

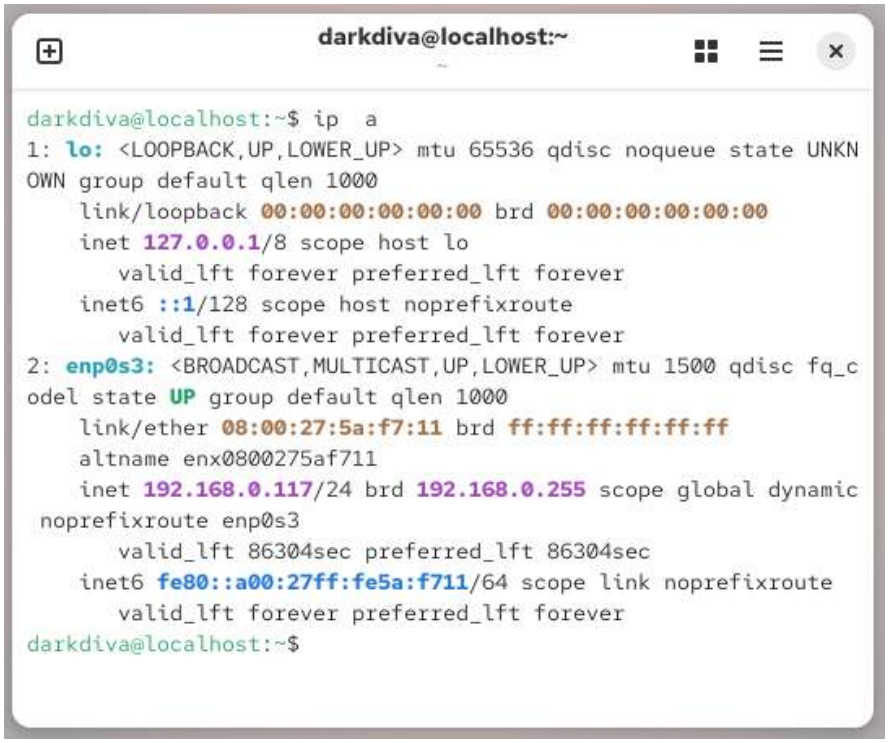


Figure 11. VM1 in Bridged mode: 192.168.0.117/24.

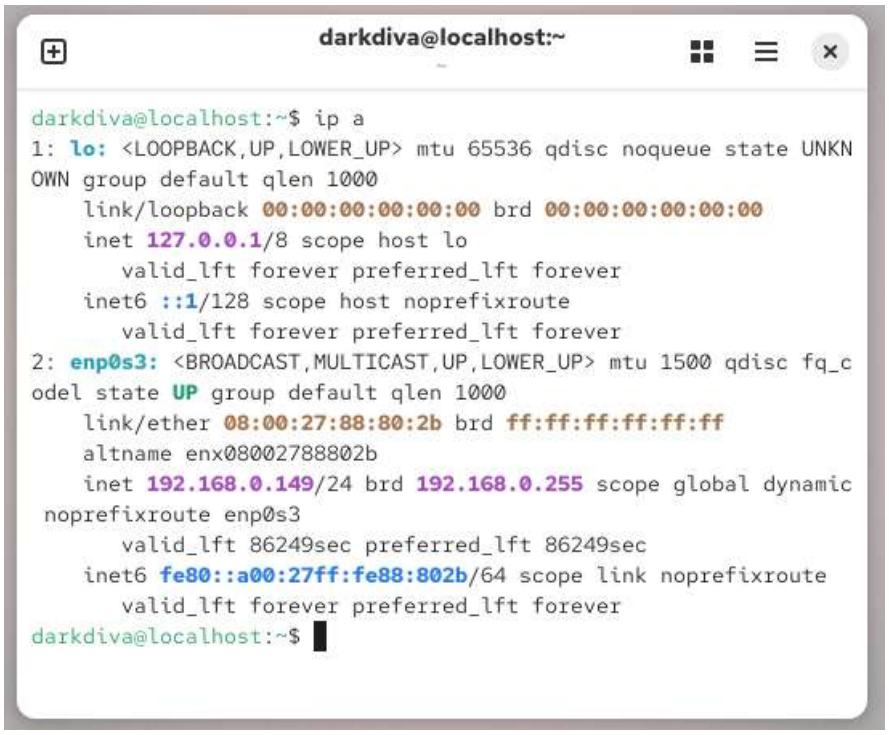


Figure 12. VM2 in Bridged mode: 192.168.0.149/24.

Table 3. Addresses in Bridged mode

Machine	IP address	Netmask
Host (Wi-Fi)	192.168.0.137	255.255.255.0
VM1	192.168.0.117	255.255.255.0 (/24)
VM2	192.168.0.149	255.255.255.0 (/24)

Ping tests

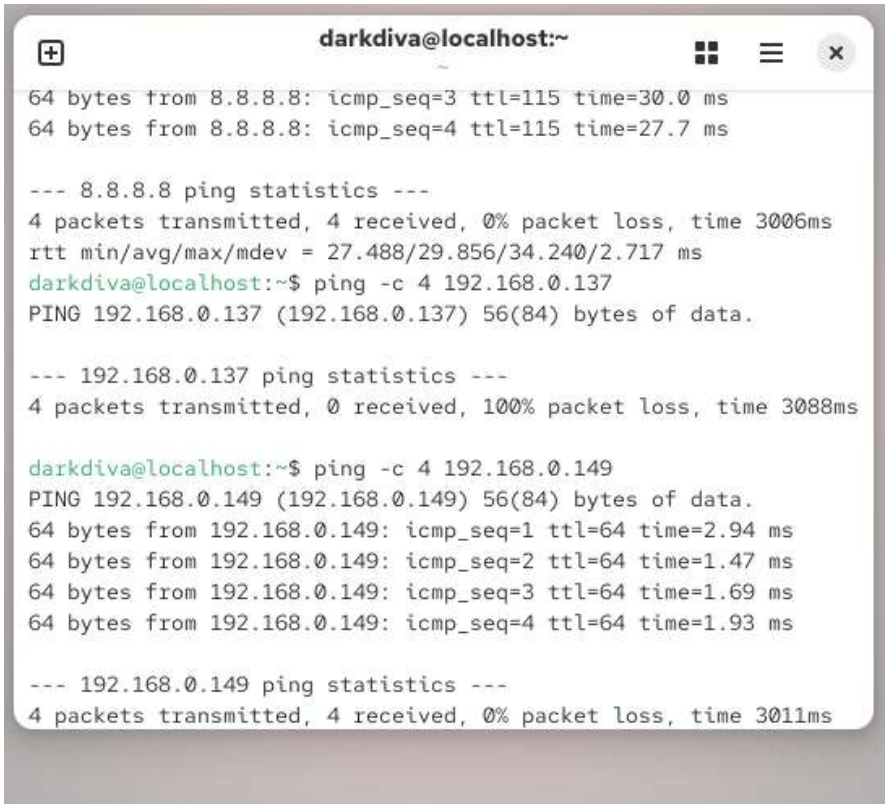


Figure 13. VM1: Internet works, host (192.168.0.137) gets 100% loss, VM2 works.

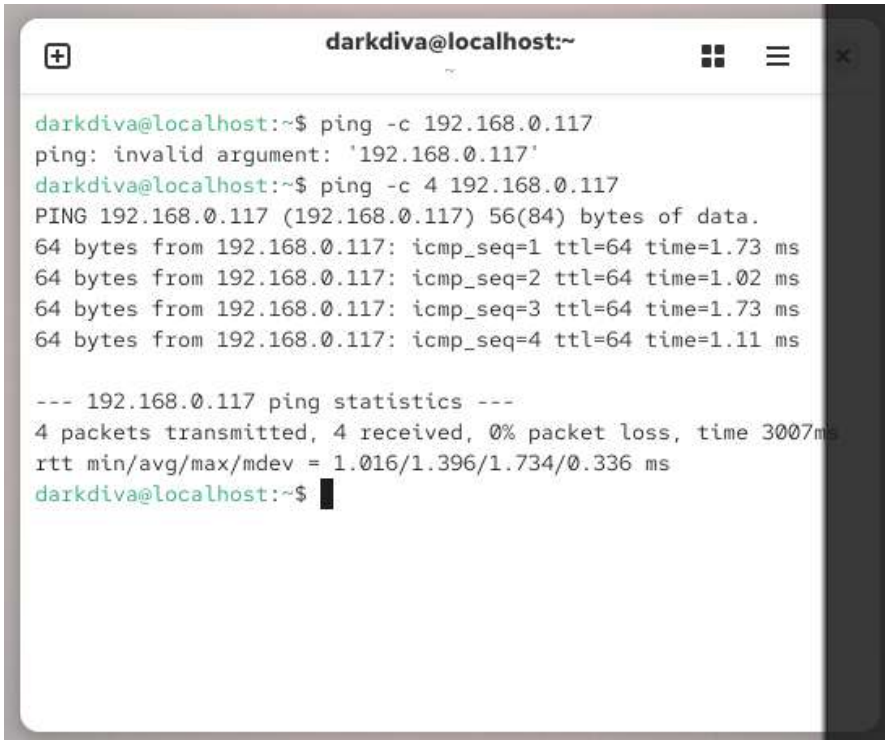


Figure 14. VM2 pings VM1 (192.168.0.117): works. The first line is a typing mistake, repeated correctly below.

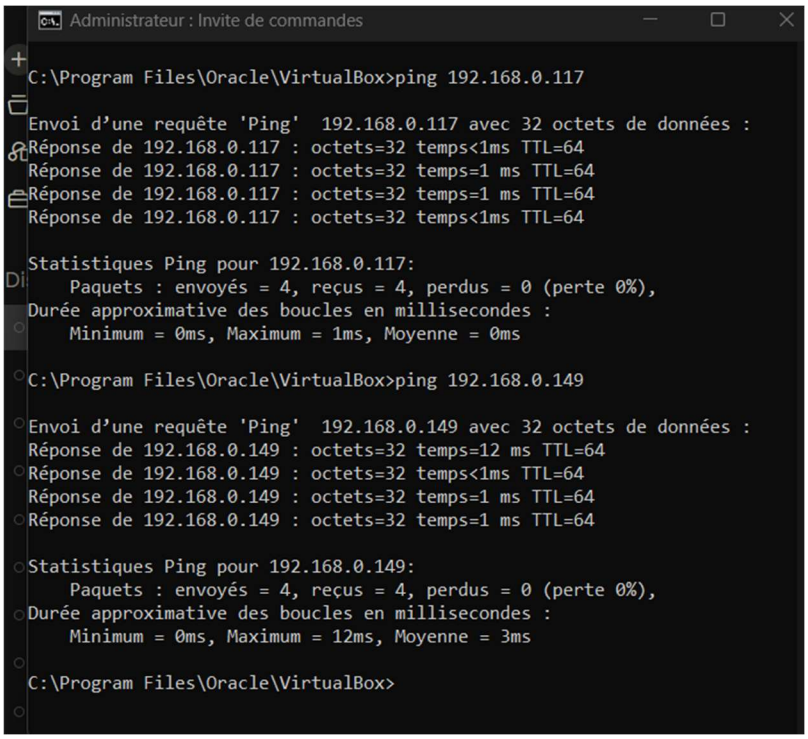


Figure 15. From Windows, both VMs answer (192.168.0.117 and 192.168.0.149).

Table 4. Connectivity in Bridged mode (VM tests made from VM1)

Source	Destination	Ping successful?
Host	VMs	Oui
VMs	Host	Non (see below)
VM1	VM2	Oui
VMs	Internet (8.8.8.8)	Oui

The VM could not ping the host

This was the one surprising result. In NAT mode the VM could ping the host, but in Bridged mode the ping to 192.168.0.137 got no reply. I made some tests to understand it.

First, VM1 can reach my router (192.168.0.1), and its ARP table contains the MAC address of my PC (shown as STALE), so the VM can find the host on the network. The ping to the host still gets 100% loss.

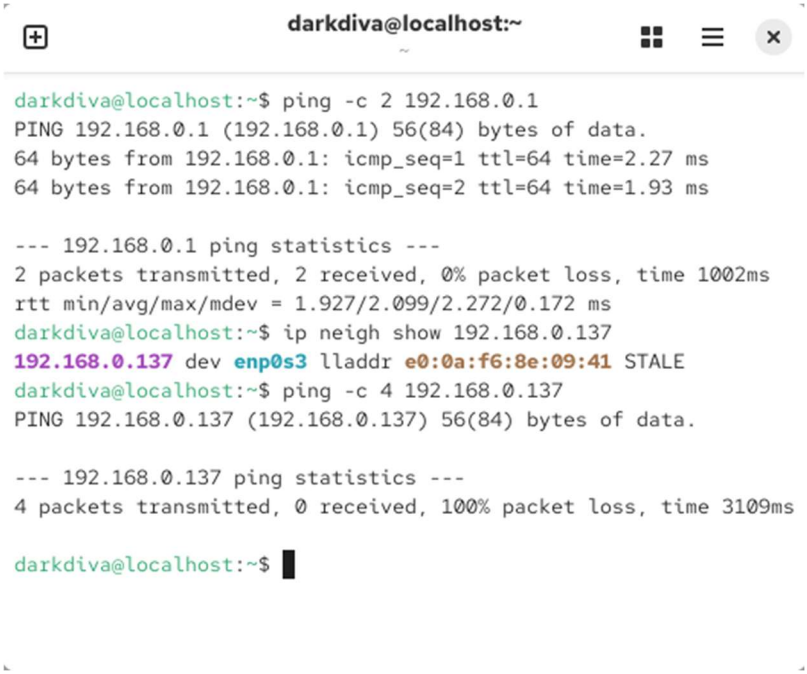


Figure 16. Router answers, the host's MAC address is known, but the host does not answer.

Second, I suspected Windows Firewall, which blocks incoming pings by default, so I added a temporary rule allowing incoming ICMP echo requests on all profiles, and tested again. The result did not change.

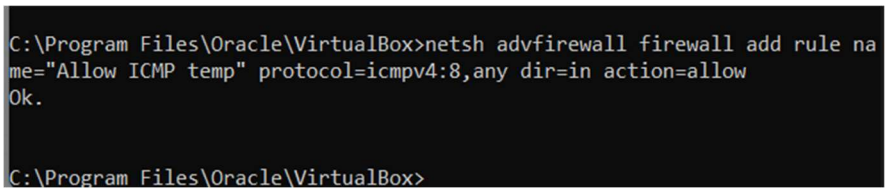


Figure 17. Adding the temporary firewall rule.

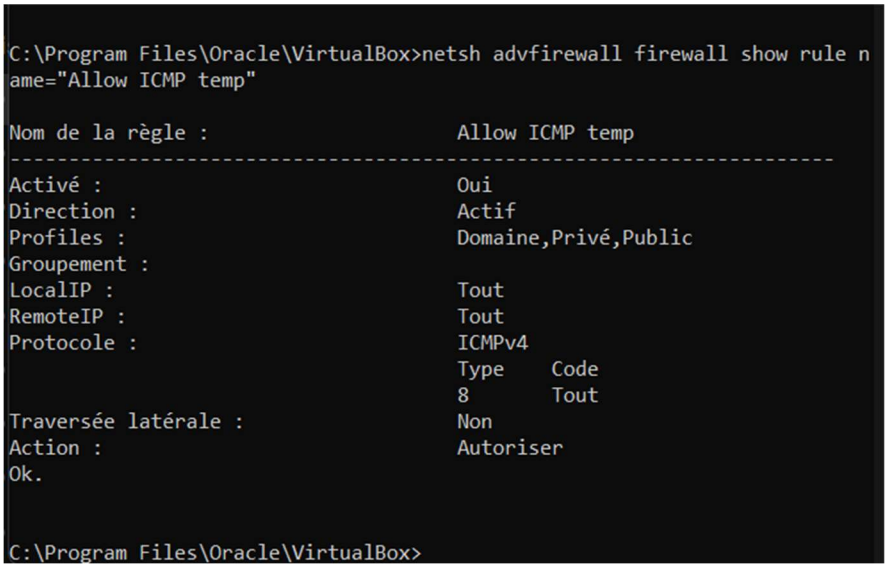


Figure 18. The rule is active on all three profiles (Domaine, Privé, Public).

```
C:\Program Files\Oracle\VirtualBox>netsh advfirewall firewall delete rule
 name="Allow ICMP temp"

1 règle(s) supprimée(s).
Ok.
```

Figure 19. The temporary rule deleted after the test.

So the firewall rule alone does not explain it, and I did not identify the real cause. It may be related to the way bridging works over a Wi-Fi adapter, but I could not prove it.

4. Internal Network mode

In this mode the VMs can only talk to other VMs that use the same network name. I shut both VMs down and set **Adapter 1** of each to **Réseau interne** with the name `intnet1`. This option was only visible after switching the settings window from **Basic** to **Expert**.

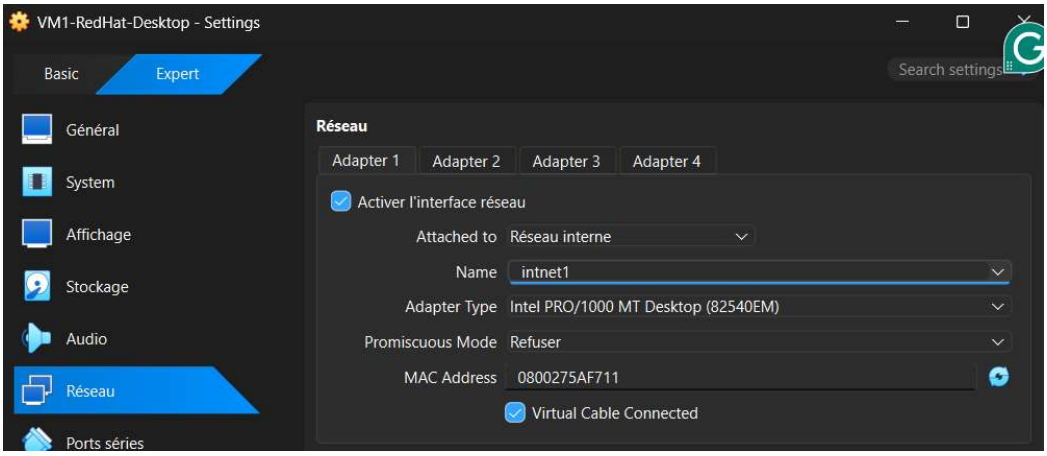


Figure 20. VM1 on the internal network intnet1.

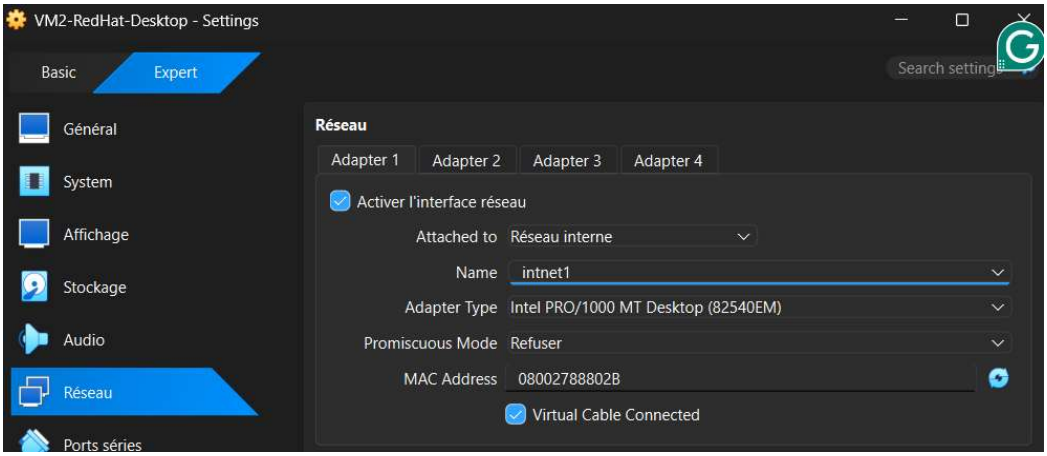


Figure 21. VM2 on the internal network intnet1.

An internal network has no DHCP server by default, so I created one on the host with `VBoxManage`. The handout uses the older option names (`--netname`, `--ip`, `--lowerip`, `--upperip`). VirtualBox 7.2.20 accepts the newer names, so I used these:

```
cd "C:\Program Files\Oracle\VirtualBox"
VBoxManage dhcpserver add --network=intnet1 --server-ip=10.10.10.1 \
    --netmask=255.255.255.0 --lower-ip=10.10.10.2 --upper-ip=10.10.10.12 --enable
```

```
VBoxManage list dhcpservers
```

The list shows the new server for `intnet1` (10.10.10.1, range 10.10.10.2 to 10.10.10.12, enabled). It also shows the server that already existed for the Host-Only adapter, which I use in section 5.

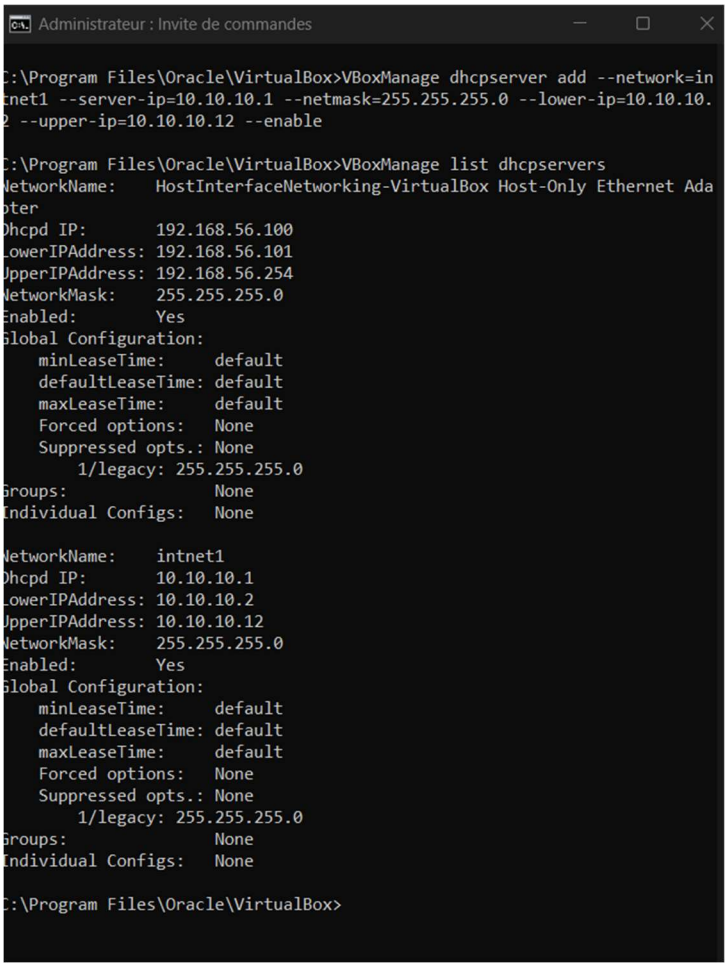


Figure 22. Creating and listing the DHCP servers.

Both VMs then received an address from that range.

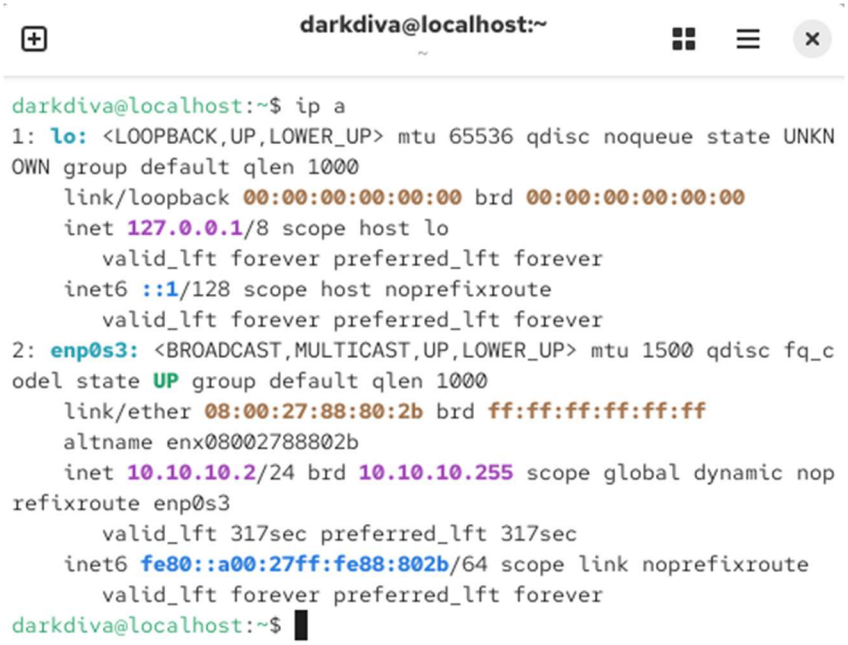


Figure 23. VM2 on intnet1: 10.10.10.2/24.

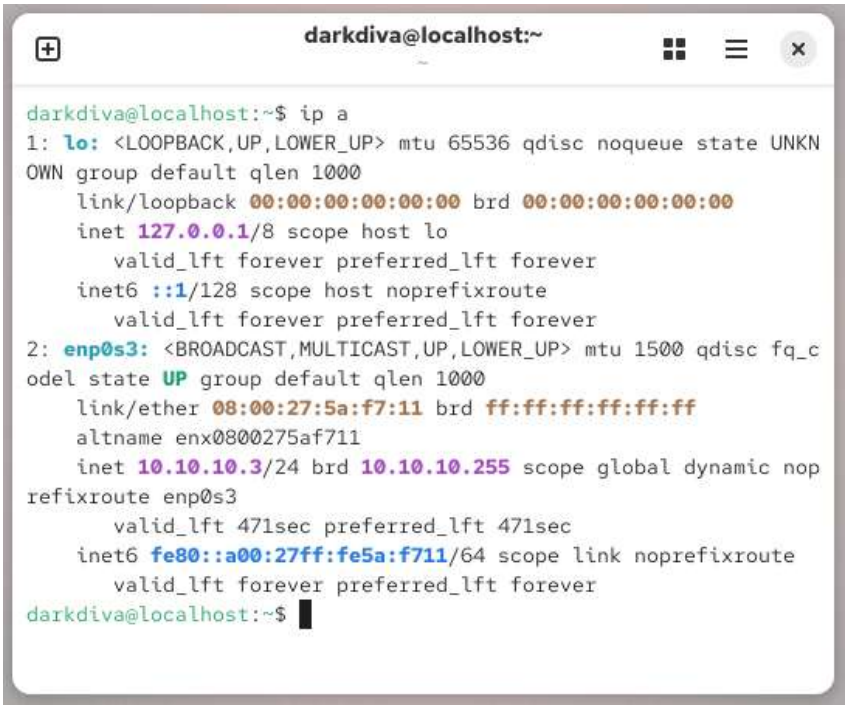


Figure 24. VM1 on intnet1: 10.10.10.3/24.

Table 5. Addresses in Internal Network mode

Machine	IP address	Netmask
Host	none (the host is not on intnet1)	n/a
VM1	10.10.10.3	255.255.255.0 (/24)
VM2	10.10.10.2	255.255.255.0 (/24)

Ping tests

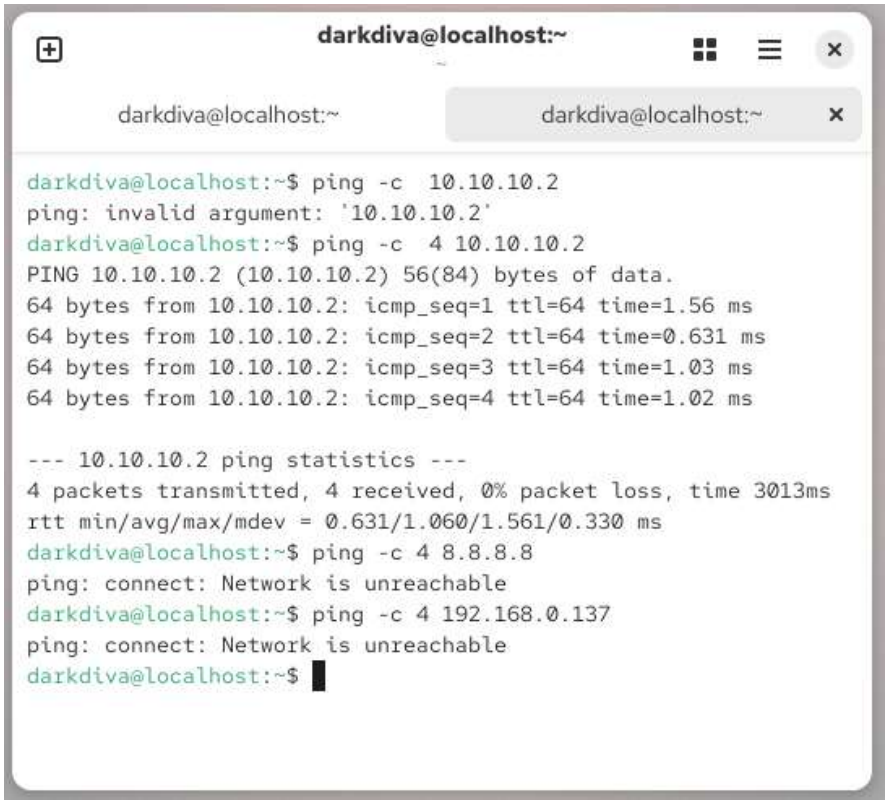


Figure 25. VM1: VM2 answers, the Internet and the host are "Network is unreachable".

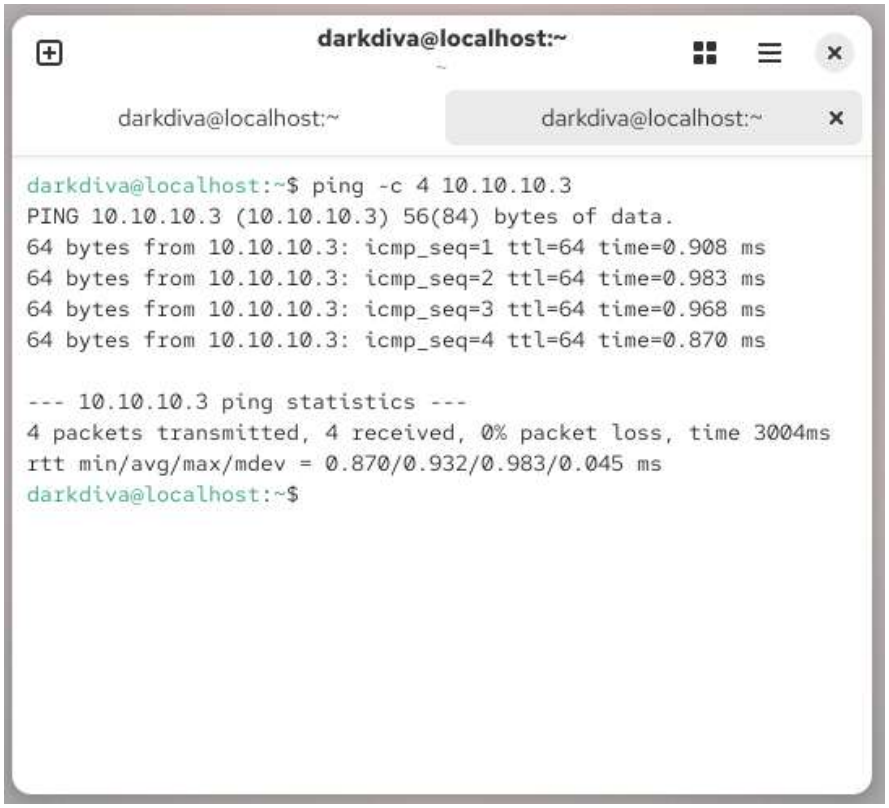


Figure 26. VM2 pings VM1 (10.10.10.3): works.

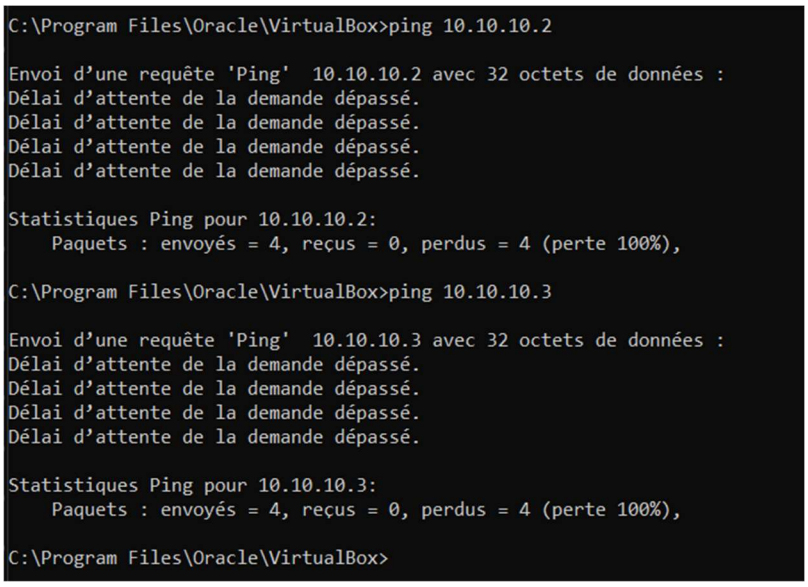


Figure 27. From Windows, both VMs time out (100% loss).

Table 6. Connectivity in Internal Network mode

Source	Destination	Ping successful?
Host	VMs	Non
VMs	Host	Non (network unreachable)
VM1	VM2	Oui
VMs	Internet (8.8.8.8)	Non (network unreachable)

Conclusion: in Internal Network mode the VMs on the same network name talk to each other, but they are cut off from the host and the Internet, because that network exists only inside VirtualBox.

5. Host-Only Adapter mode

This mode creates a closed network between the host and the VMs. VirtualBox already had a Host-Only adapter (`VirtualBox Host-Only Ethernet Adapter`, address 192.168.56.1 on the PC) with a DHCP server enabled, as the `list dhcpservers` output in section 4 shows. I shut both VMs down and set **Adapter 1** to **Réseau privé hôte** with that adapter.

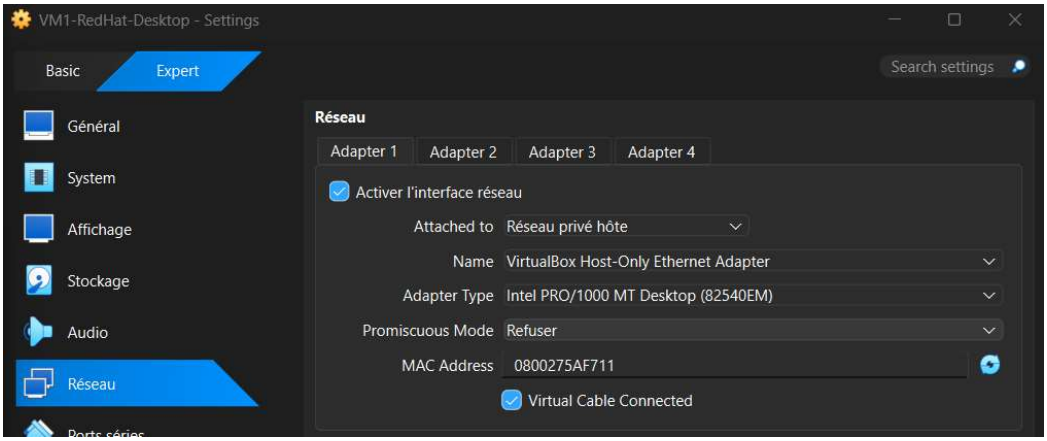


Figure 28. VM1 on the Host-Only adapter.

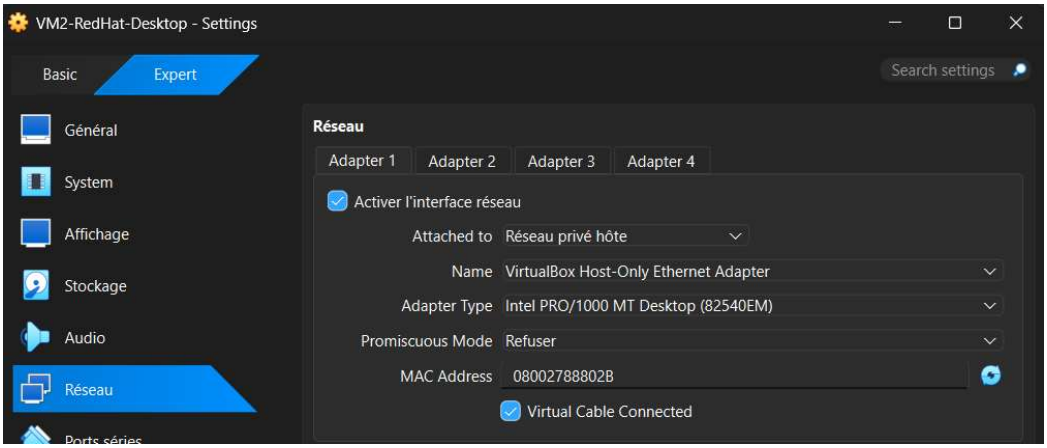


Figure 29. VM2 on the Host-Only adapter.

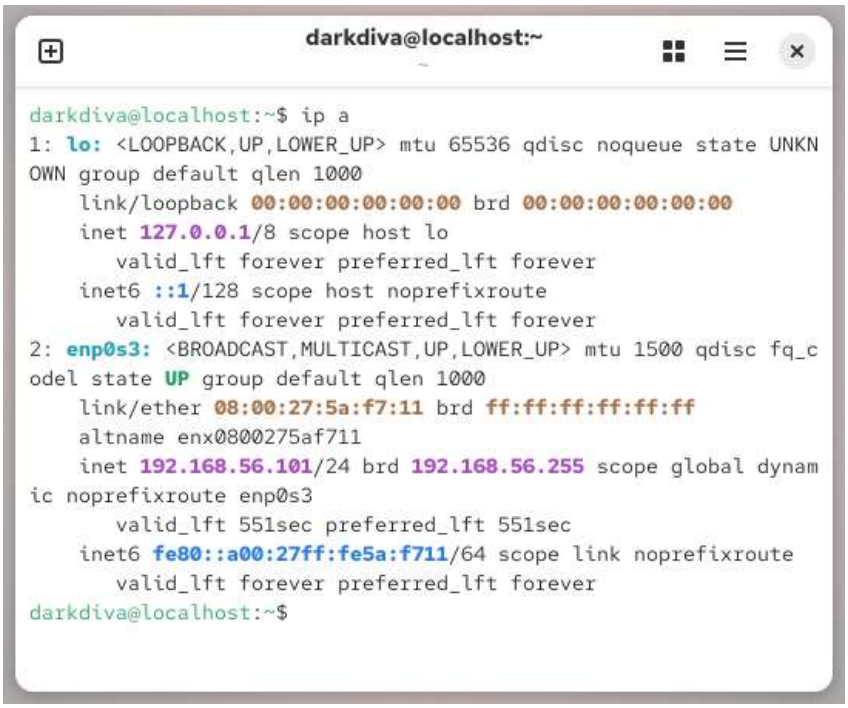


Figure 30. VM1 in Host-Only mode: 192.168.56.101/24.

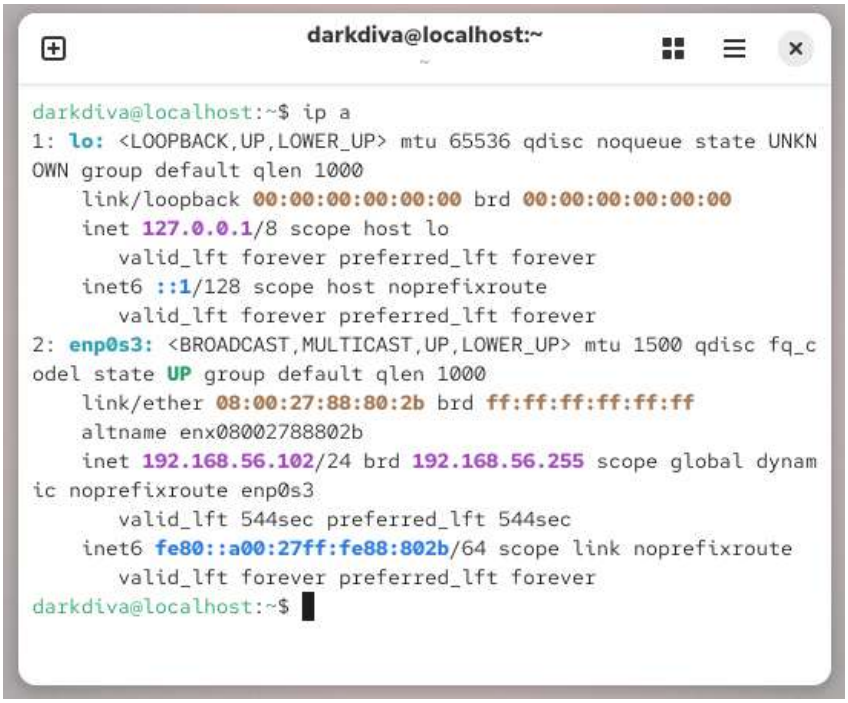


Figure 31. VM2 in Host-Only mode: 192.168.56.102/24.

Table 7. Addresses in Host-Only mode

Machine	IP address	Netmask
Host (Host-Only adapter)	192.168.56.1	255.255.255.0
VM1	192.168.56.101	255.255.255.0 (/24)
VM2	192.168.56.102	255.255.255.0 (/24)

Ping tests

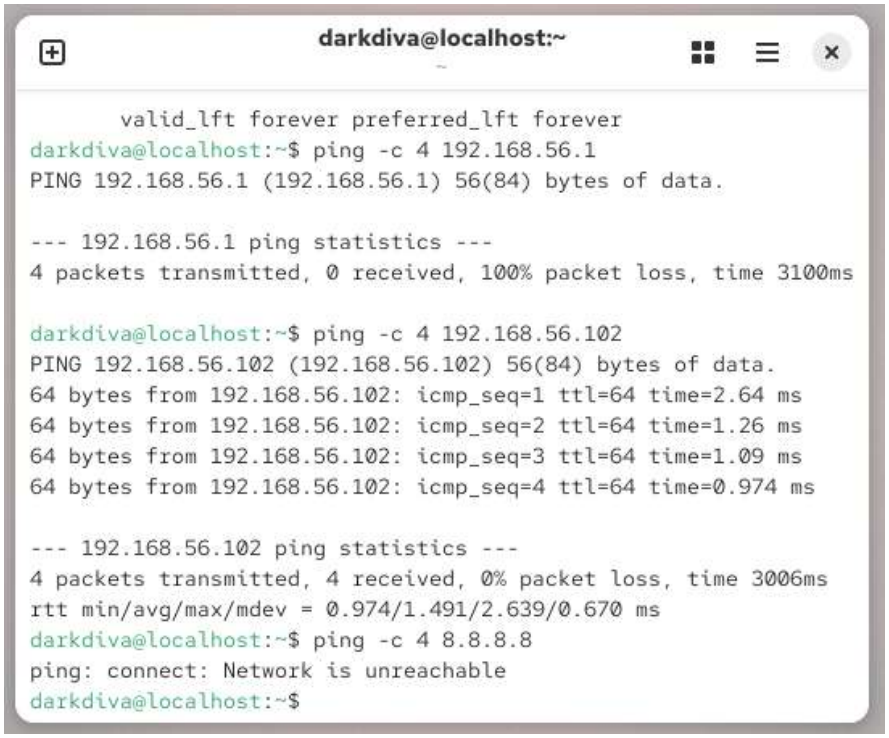


Figure 32. VM1: the host gets 100% loss, VM2 works, the Internet is unreachable.

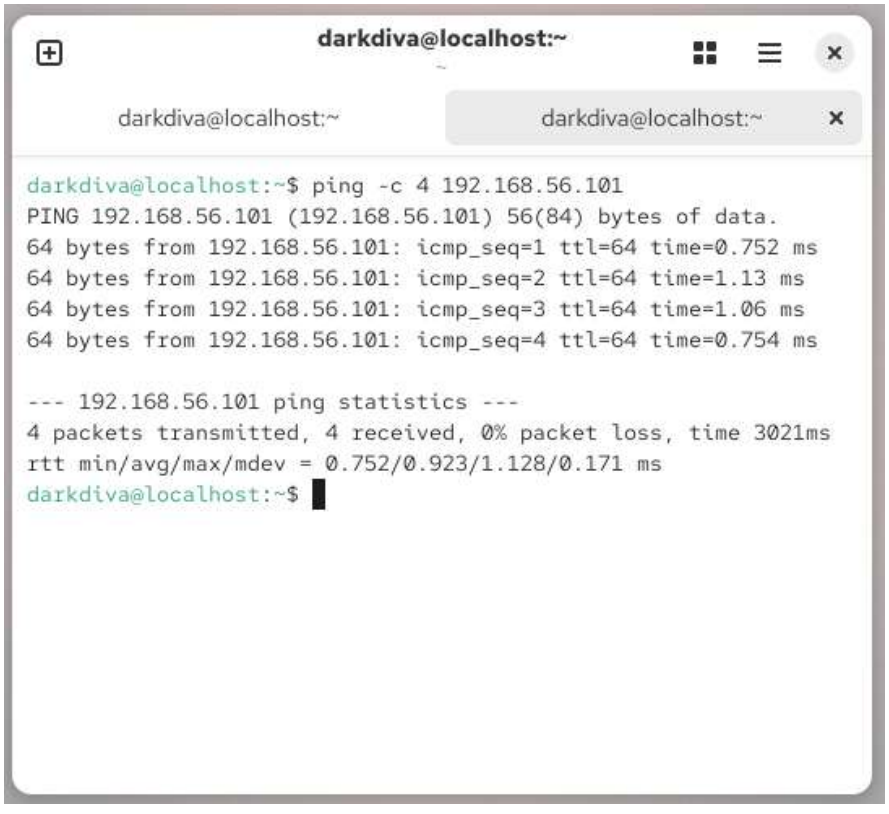


Figure 33. VM2 pings VM1 (192.168.56.101): works.

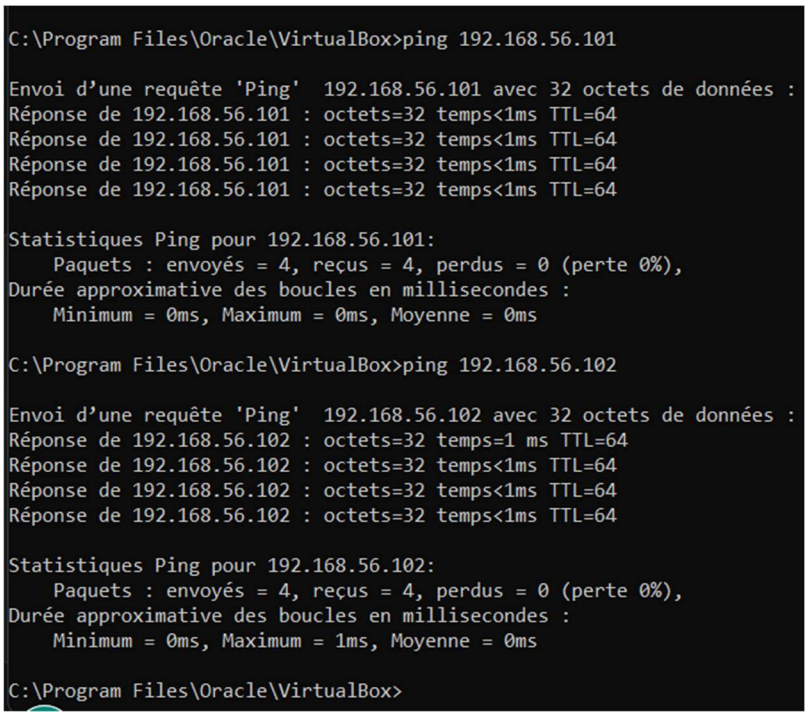


Figure 34. From Windows, both VMs answer.

Table 8. Connectivity in Host-Only mode

Source	Destination	Ping successful?
Host	VMs	Oui
VMs	Host	Non (see below)
VM1	VM2	Oui
VMs	Internet (8.8.8.8)	Non (network unreachable)

The VM could not ping the host here either (192.168.56.1), although the host could ping the VMs. I added the same temporary Windows Firewall rule as in section 3 and tested again, and nothing changed, so I did not identify the cause this time either. I removed the rule afterwards.

Conclusion: in Host-Only mode the host and the VMs see each other (apart from the unexplained VM to host ping), and there is no route to the Internet.

6. Apache and port forwarding (VM3)

Preparing VM3

I shut VM1 and VM2 down and made a full clone of VM1 called `VM3-RedHat-Desktop`. A clone keeps the settings of the original, so VM3 started in Host-Only mode, and I had to set it back to NAT.

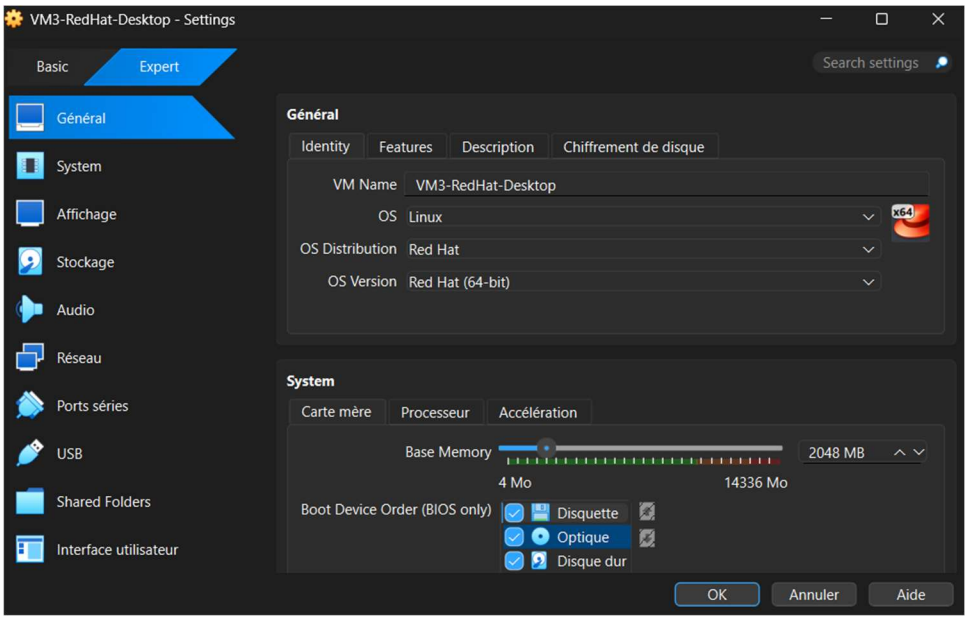


Figure 35. Settings of VM3-RedHat-Desktop after the clone.

> **Problem I ran into**
>
> VirtualBox would not let me save the settings (the OK button was grey) and showed the warning "No NAT network name currently specified". I had picked **Réseau NAT** (NAT Network) instead of plain **NAT**. They are two different modes, and the warning disappeared when I chose **NAT**.

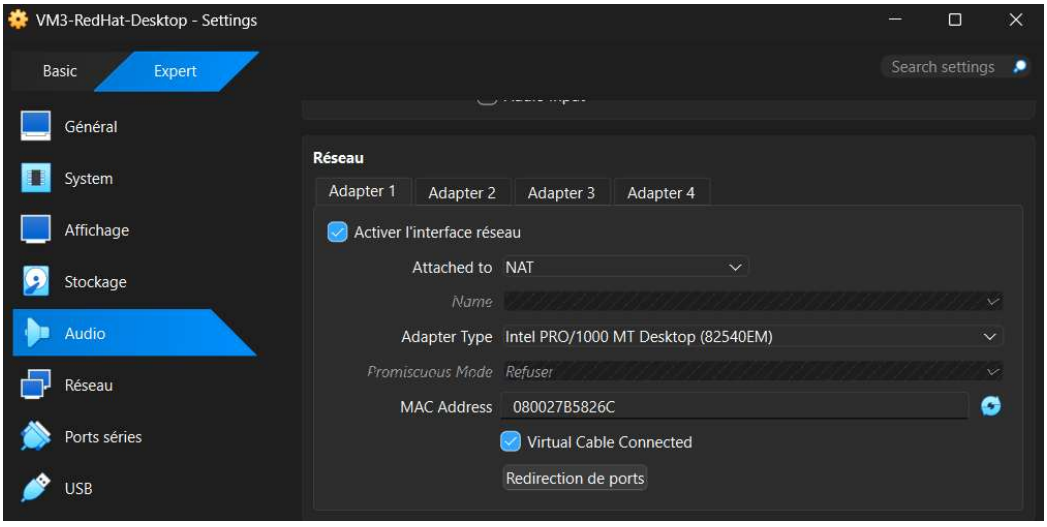


Figure 36. VM3 in NAT mode, with the Redirection de ports button available.

VM3 has Internet access and the Red Hat repositories are available, so packages can be installed:



Figure 37. Ping to 8.8.8.8 works and dnf lists the AppStream and BaseOS repositories.

Installing Apache

On Red Hat the package and the service are called `httpd` (not `apache2`):

```
sudo dnf install -y httpd
sudo systemctl enable --now httpd
sudo systemctl status httpd
```

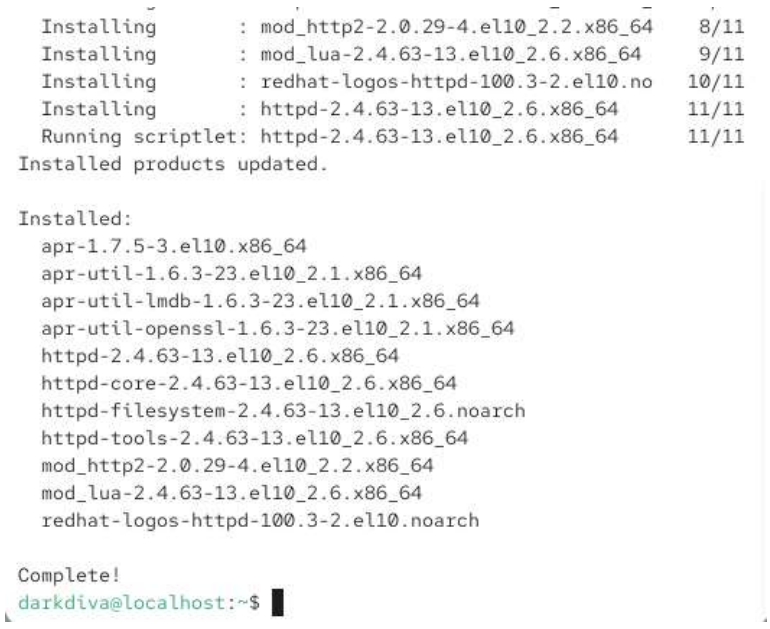


Figure 38. Installation of httpd completed.

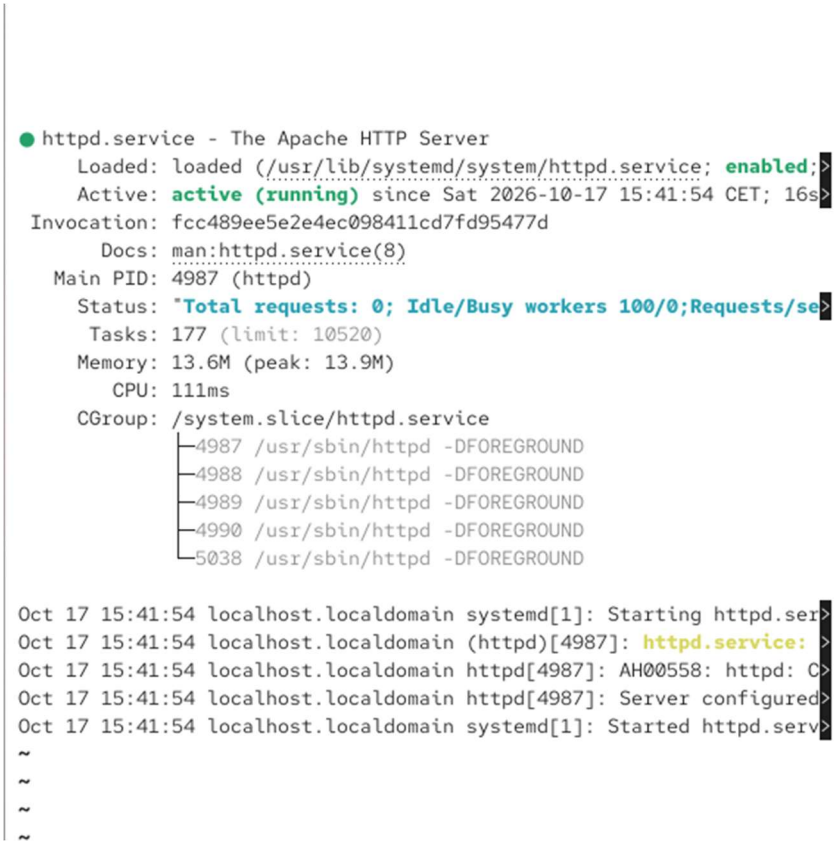


Figure 39. The httpd service is active (running) and enabled.

The handout does not mention a firewall step because Ubuntu does not block port 80 by default. Red Hat's `firewalld` does, so I opened it:

```
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload
sudo firewall-cmd --list-services
```

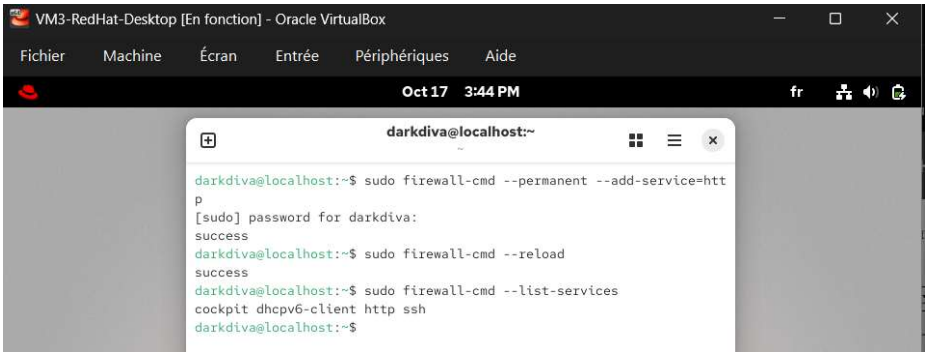


Figure 40. The http service now appears in the firewall's list.

Inside VM3, `http://localhost` shows the Red Hat Enterprise Linux Test Page, so Apache works.

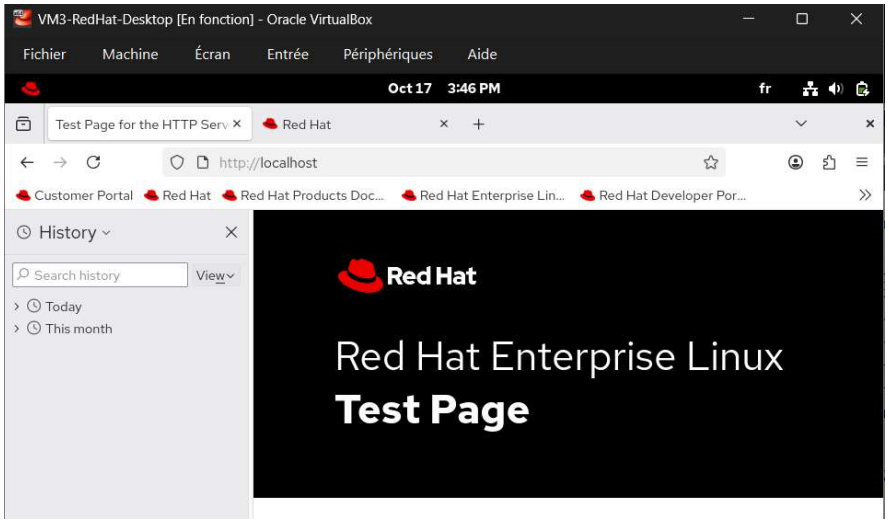


Figure 41. The Apache test page inside the VM.

Is the web server reachable from the host? (NAT mode)

VM3 is in NAT mode with the address 10.0.2.15.

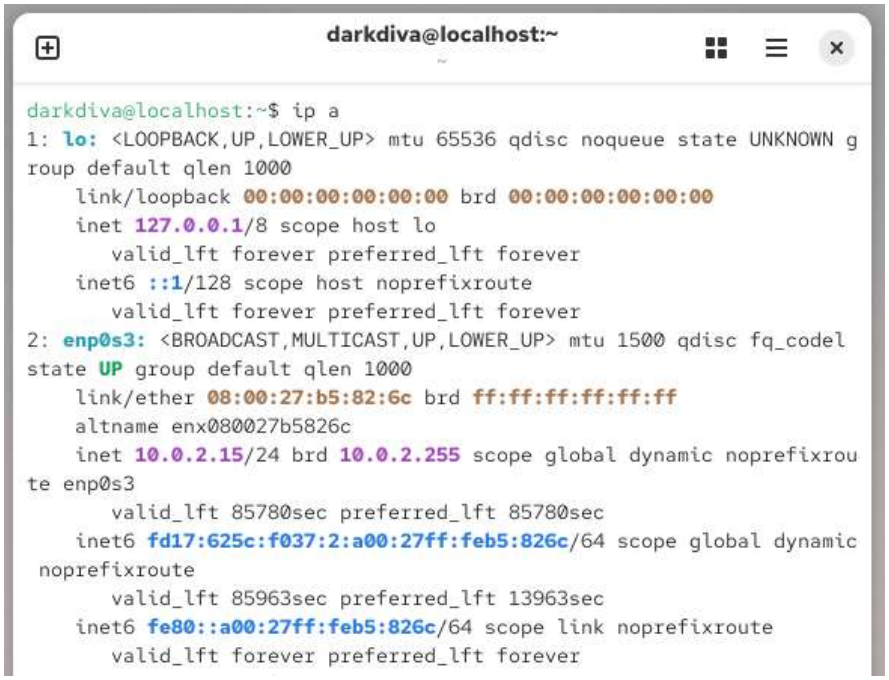


Figure 42. VM3 in NAT mode: 10.0.2.15/24.

From Chrome on Windows I tried the VM's address and then `localhost`:

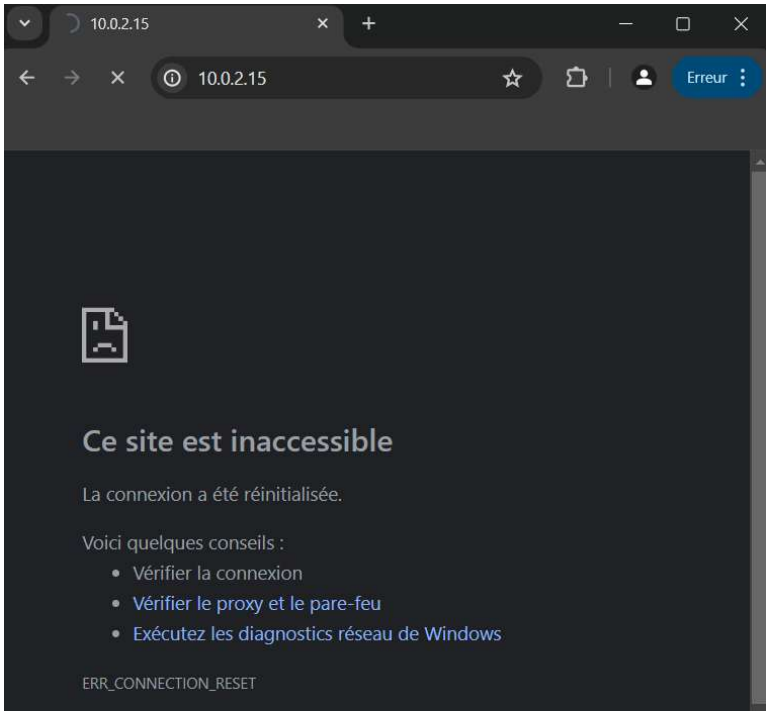


Figure 43. http://10.0.2.15 from Windows: ERR_CONNECTION_RESET.

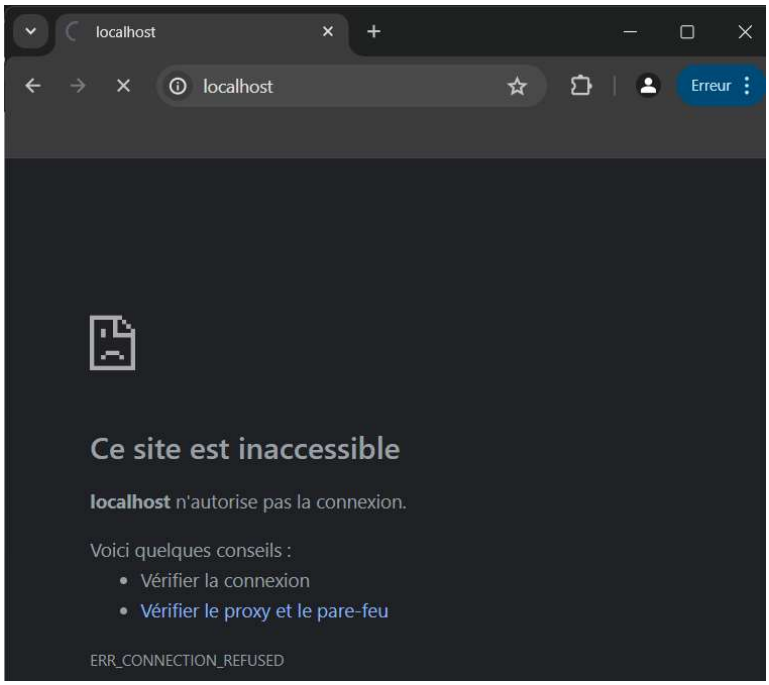


Figure 44. http://localhost from Windows: ERR_CONNECTION_REFUSED (it reaches the PC itself, not the VM).

Answer: the web server works inside the VM, but it is not reachable from the host or from outside in NAT mode.

Reaching it with Bridged mode

I changed VM3 to Bridged Adapter, restarted it and read its new address.

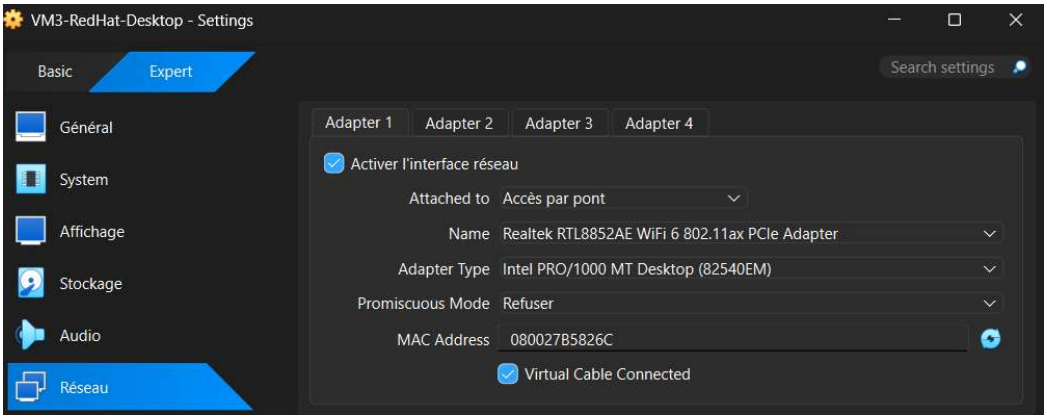


Figure 45. VM3 in Bridged mode on the Wi-Fi adapter.

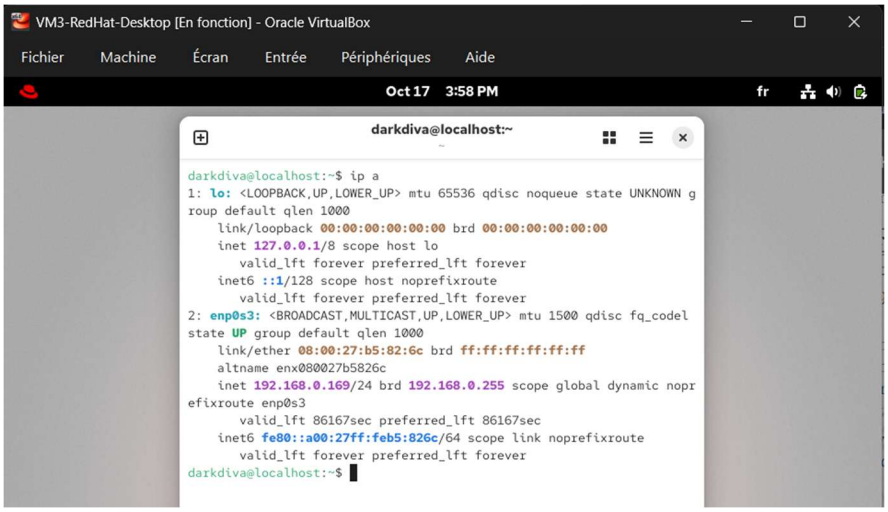


Figure 46. VM3 in Bridged mode: 192.168.0.169/24.

In Chrome on Windows I opened `http://192.168.0.169:80` (the handout writes `http//IP_VM:80`, which lacks a colon). The test page loads.

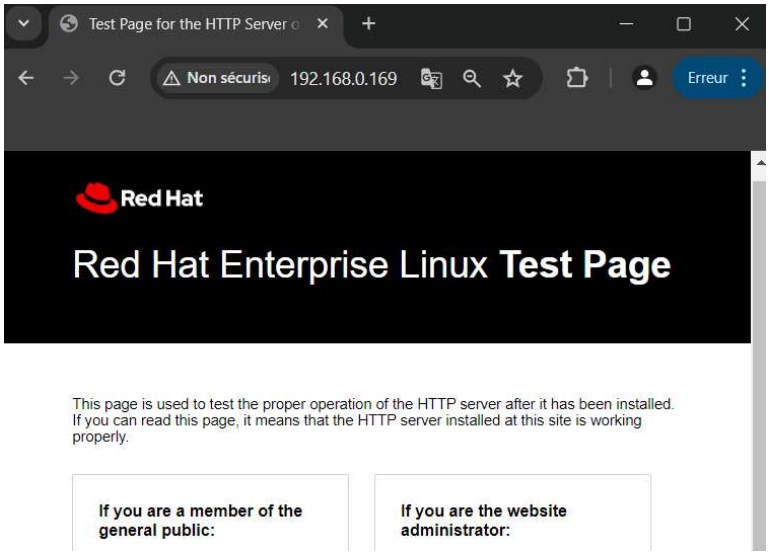


Figure 47. The Red Hat test page from Windows, in Bridged mode.

Answer: in Bridged mode the web server is reachable from the host, because the VM has its own address on the real network.

Reaching it with port forwarding (NAT mode)

The goal is to keep NAT and still reach the web server. I set VM3 back to NAT and, in the NAT settings under Advanced, added a port forwarding rule: the host port 3000 on my PC's address 192.168.0.137 goes to port 80 of the VM (10.0.2.15).

Table 9. Port forwarding rule

Name	Protocol	Host IP	Host port	Guest IP	Guest port
Rule 1	TCP	192.168.0.137	3000	10.0.2.15	80

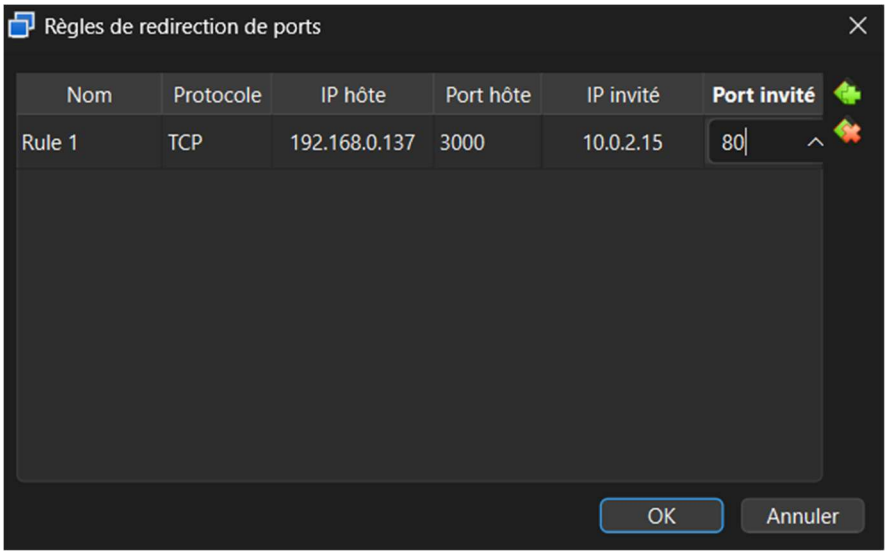


Figure 48. The port forwarding rule in VirtualBox.

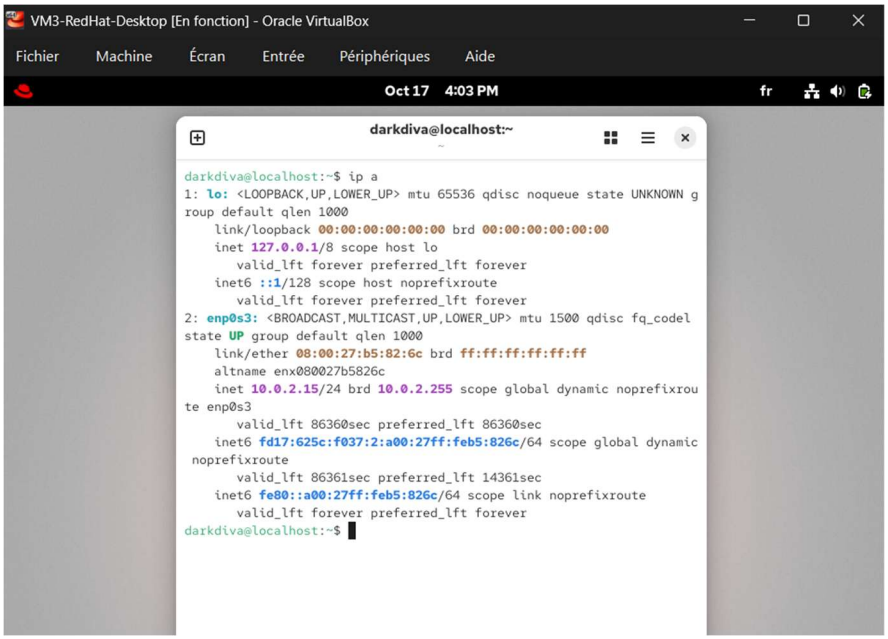


Figure 49. VM3 back in NAT mode: 10.0.2.15/24.

In Chrome on Windows I opened `http://192.168.0.137:3000`, and the Red Hat test page from the VM appeared.

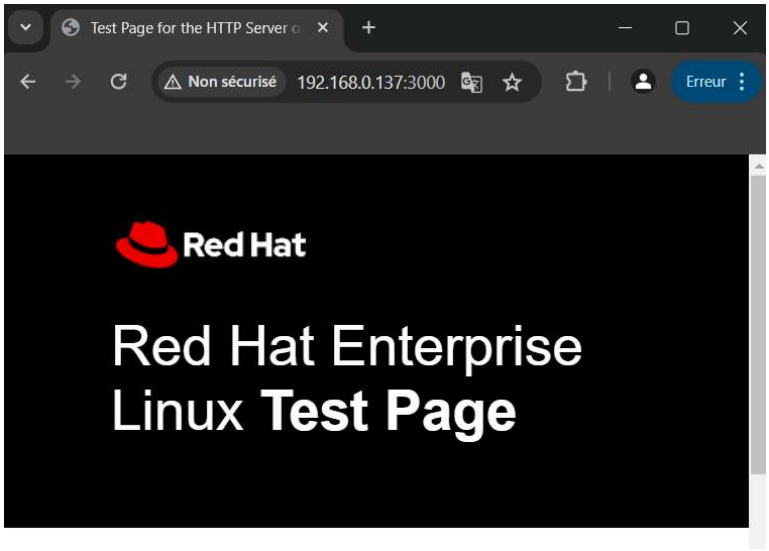


Figure 50. The test page through the forwarded port 3000.

Answer: to let the host reach the web server while keeping NAT, I can either switch to Bridged mode or use port forwarding. One note: 192.168.0.137 is the address my PC got from the Wi-Fi network, so if it changes, the rule has to be updated.

7. Comparison of the modes

Mode	VM to Internet	VM to host	VM to VM	Host to VM
NAT	Oui	Oui	Non (isolated)	Non
Bridged Adapter	Oui	Non (cause not found)	Oui	Oui
Internal Network	Non	Non	Oui	Non
Host-Only Adapter	Non	Non (cause not found)	Oui	Oui

Which mode to use depends on the goal. NAT is the simplest and gives Internet access with no setup. Bridged puts the VM on the real network, so it can be reached from other machines. Internal Network builds an isolated network between VMs. Host-Only lets the host and the VMs work together without any Internet access.

8. What I learned

The biggest lesson for me was that the same ping can mean different things in different modes. In NAT the two VMs had the same address, and a ping that looked successful was only the VM answering itself. I learned to read the address, the MAC and the delay before trusting a result.

I also saw that a clone keeps the old network settings (VM3 started in Host-Only), that an internal network needs its own DHCP server, and that on Red Hat the firewall must be opened for a web server, which Ubuntu does not require.

The two things I could not explain are the VM to host pings in Bridged and Host-Only modes. I would rather report what I tested than claim a cause I could not prove.